

Understanding controls on flow permanence in intermittent rivers to aid ecological research: integrating meteorology, geology and land cover

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ABSTRACT

Intermittent rivers, those channels that periodically cease to flow, constitute over half of the total discharge of the global river network and will likely increase in their extent owing to climatic shifts and/or water resources development. Burgeoning research on intermittent river ecology has documented the importance of the meteorologic, geologic and land-cover components of these ecosystems on structuring ecological communities, but mechanisms controlling flow permanence remain poorly understood. Here, we provide a framework of the meteorologic, geologic and land-cover controls on intermittent streamflow across different spatio-temporal scales and identify key research priorities to improve our understanding of intermittent systems so that we are better able to conserve, manage and protect them. Copyright © 2015 John Wiley & Sons, Ltd.

KEY WORDS temporary streams; flow cessation; network expansion; connectivity; continuity

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INTRODUCTION

Intermittent rivers are those fluvial landforms that periodically cease to flow and include temporary, ephemeral, seasonal and episodic streams and rivers in defined channels. The character of surface-water presence in intermittent rivers – here referred to as flow permanence – encompasses a wide range of spatio-temporal patterns in timing, frequency and duration that defines streamflow continuity through time and connectivity across space from the mesohabitat to network spatial scales (Jaeger and Olden, 2012; Jaeger *et al.*, 2014). Estimates of total length and discharge of intermittent rivers are difficult to determine, but conservative estimates suggest that intermittent rivers constitute more than 30%, and reasonably can be extended to >50%, of the total length and discharge of the global river network (Datry *et al.*, 2014a). Intermittent rivers drain all terrestrial biomes and can occur owing to natural (Figure 1a, b, c) or anthropogenic

(Figure 1d, e) drivers in all areas of the river network continuum from small headwater to large mainstem rivers (Kennard *et al.*, 2010). The prevalence of intermittent rivers is likely to increase in regions subject to water resources development and/or climate shifts (Stanley *et al.*, 1997; Rupp *et al.*, 2008; Larned *et al.*, 2010b). For example, networks with shifts towards increased aridity or snow-to-rain dominated events will likely be unable to support current or historic flows (Buttle *et al.*, 2012).

Intermittent rivers, like all rivers, possess geomorphic features that reflect the dominance of erosion by moving water that include defined banks, presence of scour and, generally, convergence of channels. However, intermittent rivers demonstrate greater variation between and within rivers in the size of their drainage areas and characteristics of their hydrology, morphology and substrate (Adams and Spotila, 2005; Tooth and Nanson, 2011; Buttle *et al.*, 2012) relative to perennial rivers (Figure 2). For example, discontinuous channels and downstream hydraulic geometry relationships that are different from perennial systems can occur as a result of downstream decreases in streamflow volume from transmission losses or the combined effect of variable discharges, erodible substrate and large amounts of bedload, particularly in drier systems (Tooth, 2000; Merritt and Wohl, 2003). Intermittency can occur in narrow, steep rivers with coarse grains (Figure 2A) as well as in wider, gradual, alluvial rivers with fine grains (Figure 2B).

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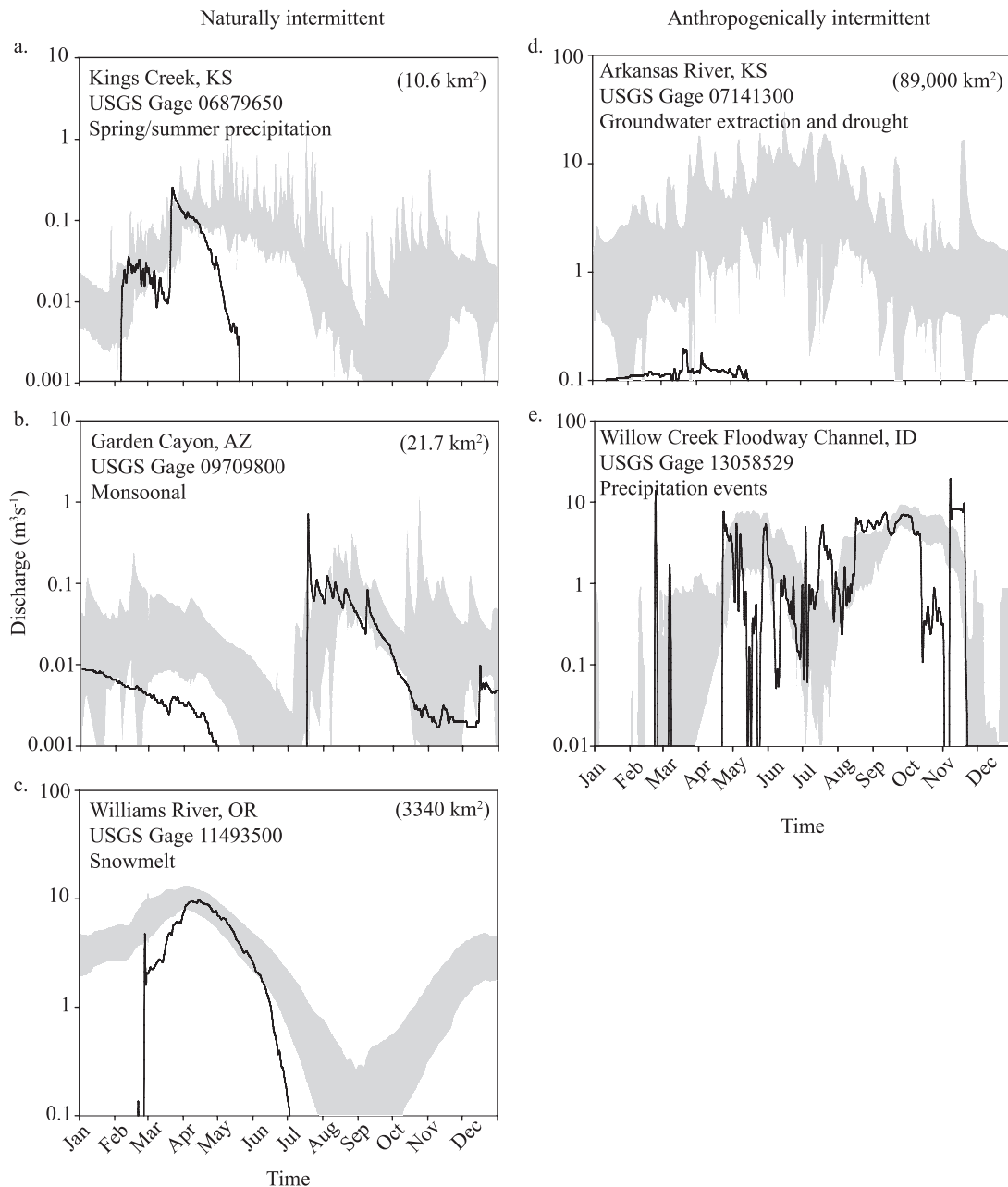


Figure 1. Examples of intermittency patterns for natural (left column) and anthropogenically altered (right column) rivers in the USA. The grey area represents the 95% confidence intervals on mean daily discharge from 1980 to 2009, and the black line represents mean daily discharge from 2012. Area is provided by the USGS but is unavailable for Willow Creek Floodway Channel.

The recognized prevalence of intermittent rivers and their expected extension due to climatic shifts and/or water resources development have resulted in a growing awareness of the ecological importance of these systems (Larned *et al.*, 2010b; Datry *et al.*, 2011; Larned *et al.*, 2011; Acuña *et al.*, 2014; Datry *et al.*, 2014a, b). Intermittent river systems undergo periodic transitions between aquatic and terrestrial phases (Figures 3 and 4), potentially enabling them to support a high level of biodiversity when both wet and dry habitats (e.g. lentic, lotic and terrestrial) are

collectively taken into account (Allen *et al.*, 2013; Datry *et al.*, 2014a). Aquatic taxa of intermittent rivers possess traits that enable them to persist in these harsh environments (e.g. high dispersal ability, rapid development and dormant stages) (Williams 2006). This typically results in lower aquatic species richness when compared with more perennial systems (Arscott *et al.*, 2010; Datry *et al.*, 2014b) supporting the notion that stream drying (i.e. the loss of surface water) is a principal driver of community structure in river networks. Terrestrial/riparian taxa have life

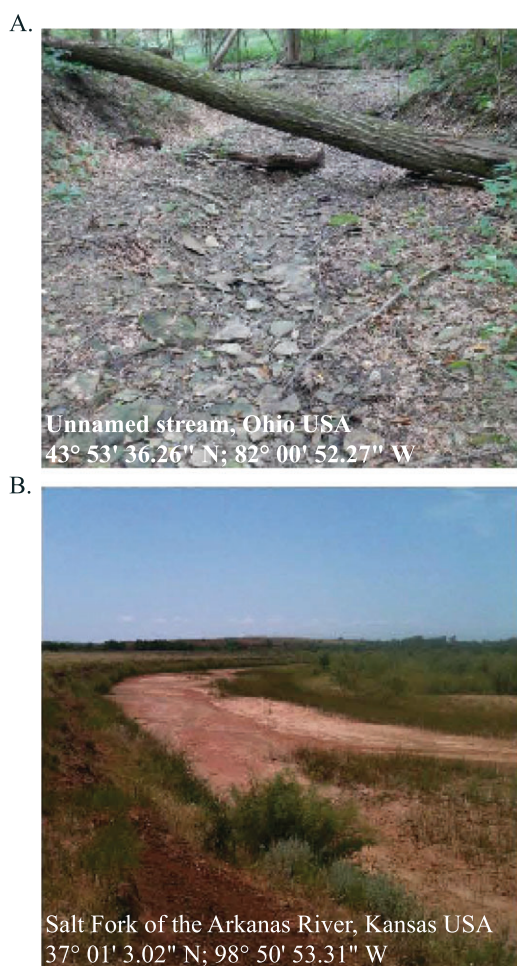


Figure 2. Examples of the highly variable morphologic forms among intermittent rivers: (A) small, steep, cobble-bed and gravel-bed river. (B) Large, gradual, sand-bed river. Approximate coordinates of where the photos were taken are given in the figure. Photograph A was taken by Katie Costigan of an unnamed river in the Killbuck River watershed, Ohio, USA. Joshua Perkin provided photograph B of the Salt Fork Arkansas River, Kansas, USA.

histories, strategies and adaptations to take advantage of limiting resources, avoid and/or endure hydrological changes, in particular inundation and current forces when flow returns to intermittent streams (Lambeets *et al.*, 2008; O'Callaghan *et al.*, 2013; McCluney and Sabo, 2014). The faunal composition of assemblages that colonize intermittent stream beds can change markedly within and between years (wet-dry cycles) and include taxa distinct from those in perennially flowing stream beds or adjacent terrestrial habitats (Stromberg *et al.*, 2009; Steward *et al.*, 2011; Dell *et al.*, 2014), but the effect of drying on adjacent riparian communities may not be as distinct (Corti and Datry, 2014). Only those taxa that have adaptations enabling them to tolerate harsh drying conditions or exploit newly available aquatic habitat (e.g. burrowing or dormant stages) can persist locally (Williams 2006). Organisms inhabiting

historically perennial rivers that become intermittent may not possess adaptations that enable them to persist in intermittent systems.

Intermittent rivers function differently from their perennial counterparts with respect to biogeochemical fluxes and may have substantial impacts on carbon and nutrient fluxes at the network, regional and global scales (Datry *et al.*, 2014a; Pisani *et al.*). The periodic transitions between aquatic and terrestrial phases in intermittent river beds, particularly immediately at their transitions, have been identified as hot spots and hot moments for biogeochemistry (McClain *et al.*, 2003) and have been further described as 'punctuated biogeochemical reactors' (Larned *et al.*, 2010b). The physiochemical changes stemming from changes in water availability, including redox state, temperature and retention, are key in controlling the sequence of biogeochemical functions (Ademollo *et al.*, 2011). During the dry phase, processing is slow, and materials accumulate and transform through photodegradation, evaporation and desorption (McLaughlin, 2008; Kerr *et al.*, 2010; Dieter *et al.*, 2013). The nature of the rewetting will largely determine the subsequent function. When flow resumes through advancing fronts, much of the stored material (e.g. leaf litter, dissolved nutrients and terrestrial organisms) is transported downstream (Obermann *et al.*, 2009). During the recession or where flow resumes gradually (e.g. upwelling groundwater), local material processing is fast (Tzoraki and Nikolaidis, 2007; Gallo *et al.*, 2013). As flow contracts, isolated pools can lead to further biogeochemical heterogeneity controlled by factors at smaller scales (Fellman *et al.*, 2013).

Despite the recognized importance of their character on river-riparian ecosystem structure and function, a fundamental and largely overlooked component of intermittent river ecology is a quantitative understanding of the controls on wetting and drying cycles. The strong longitudinal, vertical and lateral shifts in hydrologic connectivity associated with these cycles affect movement of organisms and exchange of materials and energy across aquatic habitats and aquatic-terrestrial interfaces (i.e. riparian areas). Flow attributes, including antecedent conditions, duration, magnitude, frequency and rate of change (Poff *et al.*, 1997), can be used to describe discharge patterns and spatio-temporal variations during low flow conditions. Recent ecological research has developed principles of low flow conditions that describe how these flow attributes influence river ecosystems. For example, particular characteristics of flow conditions in all rivers, and especially intermittent rivers, impact the extent and quality of available aquatic habitat, and the connectivity and exchange of materials through a river network (Jaeger *et al.*, 2014) with implications for metacommunity and metaecosystem dynamics (Larned *et al.*, 2010b; Whitney *et al.*, 2015).

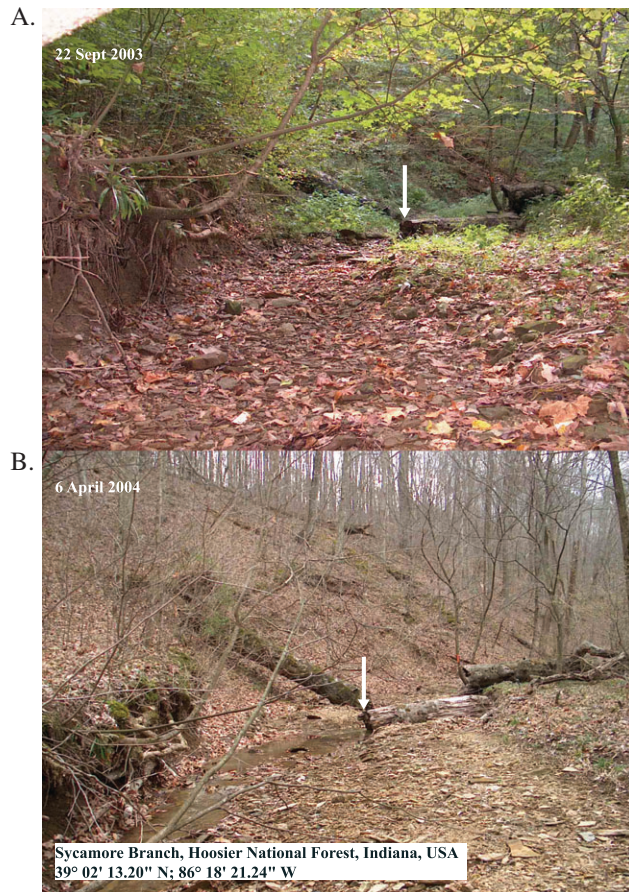


Figure 3. An example of an intermittent river during (A) terrestrial and (B) aquatic states. The white arrows in the photo mark the same location. Approximate coordinates of where the photos were taken are given in the figure. The photo was taken by Ken Fritz of Sycamore Branch, Hoosier National Forest, Indiana, USA.

Seminal work in fluvial geomorphology occurred in intermittent, dryland systems of the American Southwest (e.g. Bull, 1997; Schumm, 1979), but the emphasis did not extend to controls on flow permanence. More recent hydrologic and geologic research on intermittent rivers has largely been constrained to identifying physical indicators of intermittent rivers (Fritz *et al.*, 2008; Snelder *et al.*, 2013), with few exceptions that address mechanisms driving flow permanence (but see Godsey and Kirchner, 2014). The scarcity of research that integrates meteorologic, geologic, land cover and biologic processes (Allen *et al.*, 2014) may be attributed to the challenge of integrating different disciplines that maintain specific research methodologies, timescales and spatial extents of interest.

There is a need for ecologists to recognize the broader and more holistic freshwater and terrestrial context in which intermittent rivers function (Datry *et al.*, 2014a). This understanding requires an even broader integration beyond ecology that includes hydrologic and geologic disciplines in order to identify and quantify the mechanisms and

thresholds in hydrologic response that drive spatio-temporal patterns of flow permanence. Although there is growing recognition of the importance of intermittent rivers from scientific (e.g. Datry *et al.*, 2014a; Jaeger *et al.* 2014) and policy (e.g. the US Environmental Protection Agency's Clean Water Rule; <http://www2.epa.gov/cleanwaterrule>) perspectives, implementations of management objectives are still subject to key knowledge gaps that leave rivers vulnerable to further degradation and misuse (Acuña *et al.*, 2014). Consequently, there is an urgency to understand the mechanisms controlling flow permanence in order to predict anthropogenic impacts on intermittent river ecological processes, including their resiliency, responses to climatic shifts and other ongoing water-resource pressures (Leibowitz *et al.*, 2008; Doyle and Bernhardt, 2010). While other recent reviews include or can be extended to intermittent rivers (e.g. Tooth 2000, Smakhtin 2001; Gomi *et al.*, 2002, Levick *et al.*, 2008; McDonough *et al.*, 2011; Bracken *et al.*, 2013), the present work integrates the full suite of meteorologic, geologic and land-use processes across spatio-temporal scales within a context pertinent to biological applications. In addition, we provide research priorities for intermittent river science that will elucidate the relative controls of meteorology, geology and land cover on flow permanence.

METEOROLOGIC, GEOLOGIC AND LAND-COVER INFLUENCES ON FLOW PERMANENCE

Particularly characteristic of river networks are expansion and contraction cycles in response to variations in the amount of available water, which are most extreme for intermittent rivers, oscillating between terrestrial and aquatic states. River networks expand when moisture conditions increase and more of the network supports streamflow; networks contract when moisture reserves are depleted (Figure 4; Stanley *et al.*, 1997; Godsey and Kirchner, 2014). Figure 4 may be modified or adapted for those systems that are supported by hydrologic processes other than a fluctuating water table, like the humid region, or by upstream streamflow generation that outpaces downstream losses, like the arid region. In particular, some systems rapidly transition from completely dry to fully inundated (i.e. flash flood) where surface flows can be connected because surface-water volumes exceed losses to the subsurface. This example may not have any effect on water table elevations so may not include all steps within the illustrated sequence of wetting and drying. In addition, the patterns described here may be interrupted or initiated at any time in the sequence, as indicated by the arrows. For example, short dry spells interrupted by precipitation events can transition from contraction to expansion before a channel reaches the completely dry state.

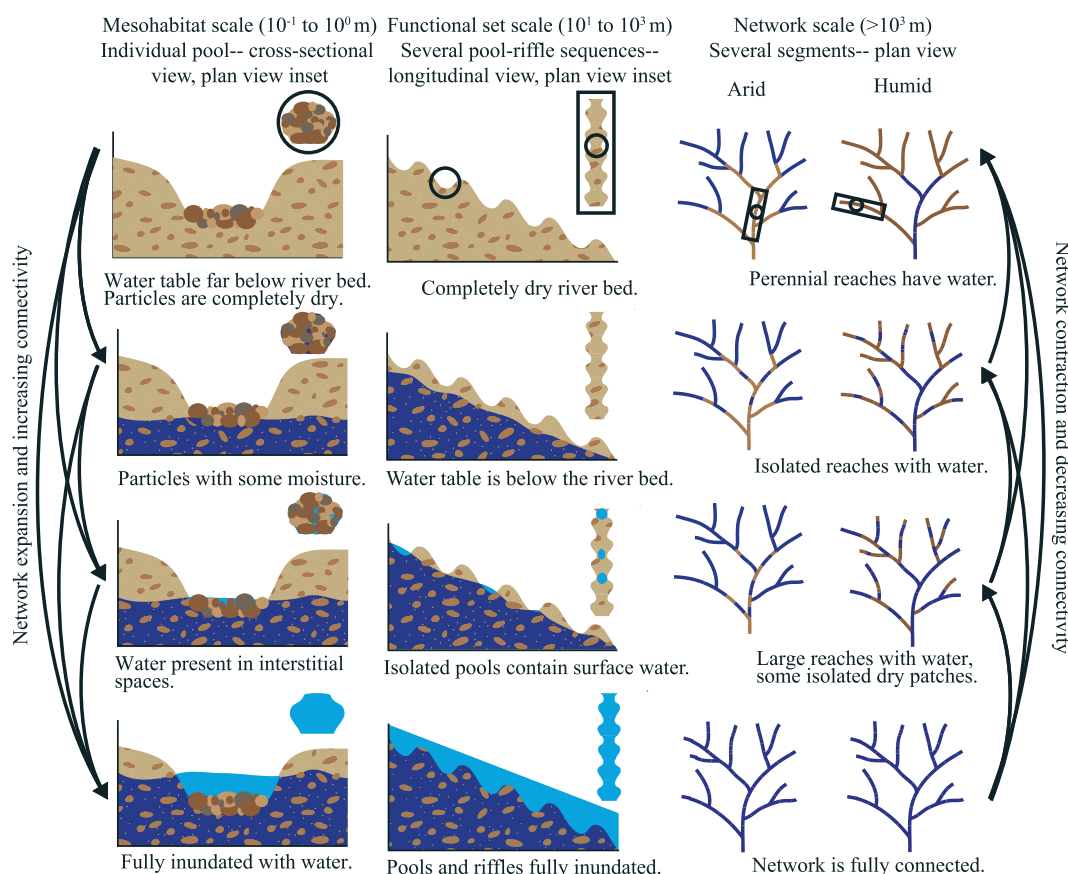


Figure 4. Expansion and contraction cycles of intermittent rivers at the mesohabitat, functional set and network scales. Flow is from left to right in the functional set and longitudinal profile and top to bottom in the network scale. The insets at the mesohabitat and functional set scale are planviews of a pool and of pool-riffle sequence, respectively. We have included the location of the insets in the network scale for arid and humid regions. This figure is adapted after Datry *et al.* (2014b), McDonough *et al.* (2011) and Stanley *et al.*, 1997.

The proximate cause of intermittent streamflow in those channels with more persistent flows is, generally, water table fluctuations relative to river bed elevation; flows occur when water tables intersect river beds and cease when the water table drops below the river bed (Konrad 2006, Larned *et al.*, 2010a, von Schiller *et al.*, 2011). Rivers may be perennially flowing with no connection to groundwater if the upstream flow generation volumes exceed losses to groundwater, which may be the case for dryland, mountainous rivers (Larned *et al.*, 2007; Larned *et al.*, 2011), but may occur in alpine, tropical and temperate regions as well (Steward *et al.*, 2012). Drying processes at all spatio-temporal scales may be interrupted at any stage by rewetting associated with precipitation and runoff (Stanley *et al.*, 1997). Network expansion and contraction cycles in intermittent river networks can range in complexity, which result in dynamic spatio-temporal gradients of flow continuity (i.e. continuous flow at a specific location through time) and flow connectivity (i.e. continuous flow across space and time) (Jaeger and Olden, 2012). The dynamic spatio-temporal patterns of flow continuity and connectivity create mosaics of aquatic and

terrestrial habitats that influence physical, chemical and biological processes in rivers (Larned *et al.*, 2010b). Those channels having streamflow generated only from overland flow or direct channel interception of precipitation will generally have lower flow permanence (Gomi *et al.*, 2002) and therefore may not experience the complexity of network expansion described earlier.

Flow permanence reflects the interactions between inputs of water and the basin conditions that mediate how water moves to and through the river network (Montgomery, 1999). River networks are nested, hierarchically organized systems (Frissell *et al.*, 1986). Patterns of streamflow are at least partially controlled by processes acting at small spatio-temporal timescales (Dodds, 1997; Fleckenstein *et al.*, 2006; Snelder *et al.*, 2013). It is important to look at the full-range spatio-temporal conditions and the interactions between spatio-temporal scales to understand patterns of streamflow. Water inputs and basin conditions, including geologic and land-cover characteristics, interact across spatio-temporal scales that range from the watershed to sub-metre and millennia to hours.

Here, we propose a framework that describes the three dominant controls on flow permanence – meteorology, geology and land cover – across spatio-temporal scales that we consider pertinent to biological processes in intermittent rivers systems (Figure 5). We introduce the framework at the broadest scale, recognizing that broad-scale structure controls finer-scale processes (Frissell *et al.*, 1986; Montgomery, 1999) that affect flow permanence. The spatio-temporal (e.g. *x*-axis) scale should be interpreted as nested hierarchical levels, and the tendency towards intermittent or perennial flow (e.g. *y*-axis) should be interpreted that those with a larger arrow have a greater influence than those with a smaller arrow. Hydrologic processes at the hillslope are applicable and can be

extended to understanding mechanisms driving flow permanence in intermittent rivers and can be coupled with conventional catchment hydrology. Therefore, we apply hillslope and catchment hydrology in addition to studies on perennial rivers and a few well-studied intermittent rivers. We build upon and expand on previous work with attention to spatio-temporal scales that transcend hydroclimatic regimes and are meaningful to biological research. We do this by applying knowledge obtained through hillslope and watershed hydrology work as the foundation to our assertions, coupled with the few well-studied intermittent rivers. Our goals are to (1) provide a conceptual framework that appropriately describes flow permanence and (2) provide a foundation to evaluate anticipated changes in

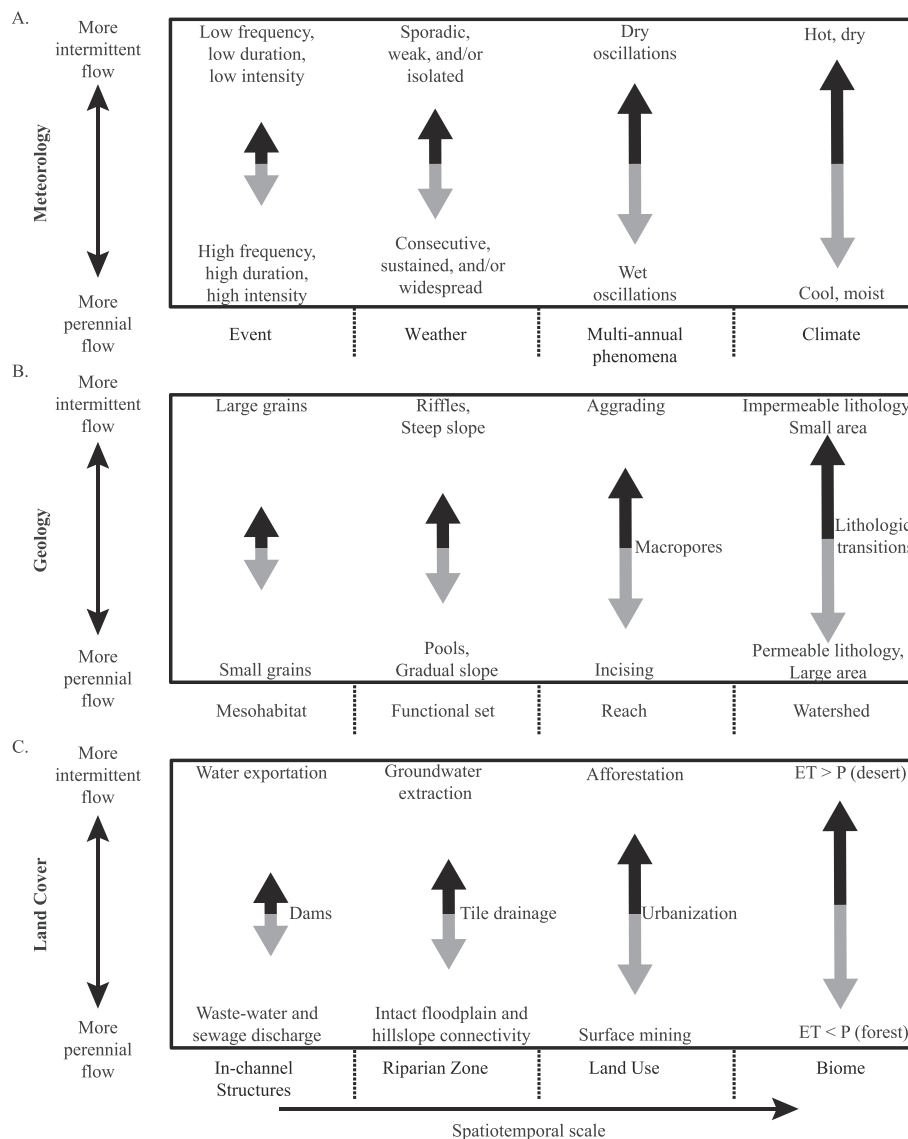


Figure 5. Spatio-temporal hierarchy of the (A) meteorologic, (B) geologic and (C) land-cover characteristics that control streamflow regimes. The direction of influence is assigned, promoting either intermittent flow (upper half with black arrows) or perennial flow (lower half with grey arrows), with the understanding that alternative cases exist as well as unknown directions of influence, which are denoted in text along the centre line of the graph where arrows radiate from. The *x*-axis is hierarchically nested. The tendency towards promoting perennial or intermittent flows is along the *y*-axis.

flow permanence under climate change and/or water-resources development scenarios.

Meteorologic

Climate, multi-annual phenomena, weather and events are the meteorologic processes that govern inputs of precipitation and energy, which are first-order controls on streamflow (Figure 5A). Climate is the longer term (>100 years) averages of environmental conditions (e.g. precipitation, temperature and humidity). Multi-annual phenomena are the inter-annual or decadal variability in environmental conditions. Weather is the inter-annual and intra-annual variations in environmental conditions. Events are unique occurrences of rain, snowstorm or meltwater at a discrete location and point in time.

Proximity of the river channel to the water table is a control in determining the spatio-temporal pattern of streamflow in intermittent rivers (Konrad, 2006; Larned *et al.*, 2010a; Von Schiller *et al.*, 2011; Goulsbra *et al.*, 2014). All of the meteorologic scales (climate, multi-annual phenomena, weather and event) presented here influence the location of the water table. Shallow water table depths occur under wetter antecedent conditions; depth to water table increases under drier antecedent conditions. While the location of the water table is a key control on streamflow, groundwater resources must be of sufficient size and have the hydraulic properties that are conducive to contribute to streamflow (Smakhtin, 2001).

The climatic regime, juxtaposed against evapotranspiration (ET) demands, is the overarching control on the occurrence of intermittent rivers. Precipitation can exceed or be exceeded by ET, depending on availability of solar radiation within the system. Arid and semi-arid biomes have low but highly variable precipitation that is often exceeded by potential ET owing to high energetic inputs that result in both a higher proportion of and higher-ordered intermittent rivers (Levick *et al.*, 2008; Snelder *et al.*, 2013). Conversely, temperate or humid regions are characterized with higher and more consistent mean annual precipitation that is not often exceeded by ET resulting in decreased frequency of and generally lower-ordered intermittent rivers. Multi-annual phenomena have oscillations that strongly influence inter-annual and intra-annual patterns in precipitation and ET. Multi-annual phenomena influence average flow permanence patterns that extend over multiple years or decades. Drought periods extending over several years may result in increased drying and reduced flow permanence. Perennial rivers may transition to intermittent rivers under prolonged and/or intense droughts (e.g. the Arkansas River at Great Bend, KS; Figure 1d). Ecological responses to potentially catastrophic disturbances (Beisner *et al.*, 2003) such as drought can include fundamental and sustained changes to community

composition (e.g. Bêche *et al.*, 2009), which in turn will influence how the broader ecosystem functions (Leberfinger *et al.*, 2010; Bruder *et al.*, 2011). Over large spatial scales, the filtering effects of drought result in more homogeneous community composition among different locations (Chase, 2007).

The finer scales of weather and event influence the immediate duration, timing and frequency of continuity and connectivity of streamflow. Weather patterns of precipitation, meltwater, ET and aquifer recharge influence streamflow and water table relationships (Konrad, 2006; Doering *et al.*, 2007). Intermittency occurs if there are sufficient precipitation deficits (Buttle *et al.*, 2012) and can manifest as changes in the spatial extent of streamflow as a consequence of decreased precipitation between years (Jaeger *et al.*, 2007). Variations in weather (e.g. environmental conditions) control meltwater generation, which will control the timing and magnitude of water delivery to the channels draining areas with snowpack and/or glaciers. The character of individual precipitation or meltwater events will determine flow continuity and connectivity at the shortest temporal scale but can affect both small and intermediate spatial scales (Jaeger and Olden, 2012). Precipitation events of higher frequencies, higher intensities and longer durations promote flow permanence compared with those of low frequency, low intensity and shorter durations. During prolonged dry periods, the channel will require longer and higher-intensity precipitation events to support streamflow compared with instances when time periods between precipitation events are short and antecedent moisture conditions are wet (Zimmermann *et al.*, 2014).

Geologic

River networks are multi-scaled, nonlinear, hierarchically nested systems that may be organized geologically, at a coarse scale, by watershed, reach, functional set and mesohabitat (Frissell *et al.*, 1986). The watershed, reach, functional set and mesohabitat scales are useful scales because they are often the scales over which various fluvial geomorphic and river ecologic research is conducted. Watersheds are the areas draining to a specific point in a river network. Reaches are large lengths of rivers with integrated geologic units with generally similar characteristics of valley confinement and channel sinuosity. Functional sets are hydromorphologic units or bedform combinations with characteristic channel dimension and hydraulic relationships (pool-riffle, step-pool, etc.) as a function of flow depth, velocity and bed material. Mesohabitats are discrete bedforms or patches within the functional set (e.g. an individual pool or riffle) with relatively homogenous flow depth, velocity and bed material. Watersheds are fixed at the broadest spatio-

temporal level, barring instances of low-frequency, extreme magnitude events (e.g. tectonic uplift, volcanism and sudden climatic shifts), whereas, at the finest level, mesohabitats are highly dynamic.

Lithology governs the types of rock and soils that develop at the watershed scale, which exerts critical controls on hillslope water movement, to and through the subsurface, and its eventual delivery to the river. Permeable soils and preferential flowpaths facilitate rapid water delivery to channels and rapid expansion and contraction cycles (Doering *et al.*, 2007). Highly permeable surficial geology and soils with rapid infiltration enhance flow permanence whereby water infiltrates during wet periods and re-emerges to contribute to streamflow during drier periods (Jencso and McGlynn, 2011). Impermeable bedrock mantled by a thin layer of permeable surficial geology and soil results in insufficient storage to sustain flows and thus reduced flow permanence. Transitions in surficial geology alter subsurface storage capabilities (Payn *et al.*, 2009) that can either promote or reduce hydrologic connectivity and flow permanence (Jencso and McGlynn, 2011). Basin shape will also influence flow permanence. For example, small, elongated watersheds with steep slopes tend to more often support intermittent flows as a consequence of rapid water delivery to (e.g. short flowpath distances) and processing through rivers (Snelder *et al.*, 2013).

Local slope influences river–aquifer exchanges, which promotes flow intermittency or permanence at the reach and functional set scales (Goulsbra *et al.*, 2014). Incising rivers with slopes steeper than the regional groundwater table slope induce hydraulic gradients that drive groundwater towards the river (Konrad, 2006). Aggrading reaches with river bed elevations above the regional water table produce hydraulic gradients that force surface water from the river into the aquifer (Konrad, 2006), promoting drying and decreased flow permanence. Preferential flow paths are, potentially, key controls on flow permanence (Payn *et al.*, 2012) and the spatio-temporal variability in water sources (Payn *et al.*, 2009). Springs and seeps that flow along fractures can serve as dominant and consistent sources of streamflow (Payn *et al.*, 2012), which promotes flow permanence. Soil macropores are preferential flow paths that develop by action of soil fauna and plant roots, physical or chemical weathering of materials, and/or by subsurface flows or piping (Beven and Germann, 1982). Macropores can temporarily but fundamentally alter the hillslope–river hydrologic relationships by producing a desynchronization between streamflow and precipitation that will vary inter-annually and intra-annually and can promote or diminish flow permanence (Zimmermann *et al.*, 2014; Costigan *et al.*, 2015).

River bed permeability is a primary control on flow permanence at the functional set and mesohabitat scales.

The residence time of water in intermittent rivers is brief over permeable substrates (Bull, 1997) as a consequence of higher transmission losses to groundwater (Levick *et al.*, 2008). The relative importance of transmission losses in governing streamflow is largely unknown (Smakhtin, 2001) but likely plays a substantial role in the persistence of local surface water (Buttle *et al.*, 2012). Local hydraulic gradients attributed to local changes in bed slope variations occur within and between mesohabitats (e.g. pool, riffle and run) and will drive the movement of surface water into and out of rivers (Stanley *et al.*, 1997; Konrad, 2006). For example, streamflow goes subsurface at the downstream extent of pools and upstream extent of riffles and resurfaces at the downstream extent of riffles. At the mesohabitat scale, as water vacates the channel, the tops of rocks are the first to dry and the last to become wet when the water returns to the channel.

Land cover

Land cover is the material that covers the surface of the Earth and reflects the interactions between meteorologic, geologic, ecologic and anthropogenic activities on that landscape. We organize land cover as a function of distance from the river into biome, land use, riparian zone and in-channel engineered structures (Figure 5C). At the broadest levels, climate and lithology regulate the natural biome of the region. The natural biome of a region will, in part, determine the particular land uses that may occur, the composition and structure of riparian zones that can include engineering activities, and the types of engineered structures present within the river. Land use is the anthropogenic management and modification of natural biomes. Riparian zones are the interface between terrestrial and aquatic systems, which can include substantial anthropogenic activities such as levee construction, mining and other activities across the floodplain and adjacent terraces and hillslopes. In-channel engineered infrastructure involves direct river manipulations (e.g. dams, culverts and weirs).

Intermittency occurs in all terrestrial biomes but is more likely to occur in dryland areas such as grasslands (Figure 1a), deserts (Figure 1b) and tundra (Poff, 1996; Dodds, 1997). The agent, severity and frequency of land-cover disturbances, both natural and anthropogenic, will influence flow permanence at the biome, land use and riparian zone levels. Understanding these natural as well as anthropogenic influences is critical to understand how human-related activities will change these systems. Land-cover disturbances influence streamflow (or change hydrologic regime) by altering infiltration capacity, ET and canopy interception of water. Intermittent headwaters have, in particular, been extensively disturbed by anthropogenic activities such as being filled or diverted to support

development activities (Roy *et al.*, 2009). Large-scale timber harvesting or large stand-replacing wildfires will lower ET losses, which promote perennial flow, whereas large-scale afforestation will raise ET losses, which promote intermittent flows. Impervious surfaces from urban land use can decrease flow permanence by reducing groundwater recharge (owing to reduced infiltration capacity); however, leaky infrastructure and increased river bed incision as a consequence of alterations in runoff and streamflow patterns may sustain surface flows in urban intermittent rivers (Lerner, 2002; Fritz *et al.*, 2013). For example in the southwestern USA, the duration and number of flow events in urbanized areas were greater than in less disturbed areas, which was attributed to reductions in infiltration capacities and subsequent increases in runoff events within the watershed (Gungle, 2006). Urbanization causes increases in impervious surface covers that reduce groundwater recharge and infiltration, which has resulted in intermittent rivers having particularly flashy hydrographs (Rheinhardt *et al.*, 2009). Water is routed to and through the river more rapidly when impervious surfaces comprise riparian zones or roads intersecting the river. Soil compaction associated with large-scale anthropogenic activities (e.g. surface and subsurface mining, and heavy machinery traffic) changes the natural meteorologic and geologic structure and results in rapid delivery of water to rivers, which can either promote or reduce flow permanence (Peirce and Lindsay, 2015). Groundwater extraction promotes intermittency by depleting aquifers.

Water resource development at the land use and riparian zone scales alter streamflows. Perennial rivers may transition to an intermittent regime if there is substantial drawdown of an aquifer (Falke *et al.*, 2011). For instance, the Arkansas River, which flows across the Ogallala Aquifer, has sections that have recently gone dry (Figure 1d) as a result of extensive and intensive irrigation practices (Falke *et al.*, 2011). Anthropogenic delivery systems (e.g. tile drainage systems, sewers and irrigation systems) can usurp catchment characteristics in controlling the patterns of water delivery to rivers (Helton *et al.*, 2010). A common agricultural practice is artificial subsurface drainages systems (e.g. tile drainage) that remove near-surface soil moisture that can rapidly deliver water to the channel (Schilling and Helmers, 2008; Klaus *et al.*, 2013).

In-channel structures, whether natural (e.g. large wood) or engineered, mediate whether or not water is able to persist in the river at the smallest spatio-temporal scales. Direct importation or exportation of water via inter-basin transfers can substantially restructure natural flow regimes (Larned *et al.*, 2010b; Steward *et al.*, 2012). Diversion structures, weirs and dams intercept flows and inundate upstream sections and may cause downstream portions of streams to become intermittent (Caruso and Haynes, 2011; Steward *et al.*, 2012). In cases where dam releases are

consistent and constant or when water is discharged from mining or sewage treatment facilities, intermittent rivers may transition to perennial (Hassan and Egozi, 2001; Steward *et al.*, 2012).

RESEARCH PRIORITIES

Understanding meteorologic, geologic and land-cover controls of intermittency has significant implications for the ecology and management of intermittent networks (Allen *et al.*, 2014). Here, we identify critical knowledge gaps in our understanding of the controls of flow permanence and provide practical and field-based approaches to strengthen insights on the relative importance of meteorologic, geologic and land-cover controls on intermittency by linking information across multiple spatio-temporal scales. Many of the controls on flow permanence were inferred from the literature of perennial river systems; intermittent river-targeted research is necessary to test and validate these assertions.

There have been numerous studies on hillslope and watershed hydrology. The key to understanding intermittent streams is to identify the threshold behaviour of channelized surface flow generation and the patterns in space and time of when that threshold is crossed. Future studies investigating the influence of surface-groundwater interactions on flow permanence are especially important because transmission losses, infiltration and effluence are primary controls on streamflow in those intermittent rivers influenced by water table fluctuations (Shanafi and Cook, 2014). Applying what we know about boundary conditions from streamflow generation (i.e. hillslope/watershed hydrology) studies to new research on alluvial systems that focuses on understanding the threshold behaviour of surface-water flow in the channel is necessary in order to gain new insight into predicting the spatio-temporal patterns of intermittent flow. Future research addressing the timing and pathways of water within intermittent rivers will be of particular importance.

Effects of urban land use on flow permanence represent another area of needed research given the extent and continued increase of urbanization and the confounding interactions and inconsistent directionality of streamflow changes in response to urbanization (Fritz *et al.*, 2013). Lag times may hinder the ability to detect the effects of recent land-use change on river morphology because intermittent rivers may only flow briefly (Rendell and Alexander, 1979; Benda *et al.*, 2005; Miller and Doyle, 2014). Concurrent research on the influence of natural disturbances on flow permanence is also necessary to disentangle the complexity associated with the effects of human land use, particularly as water resources are managed more intensively. With projected growing demand on water resources, other

land-use activities such as inter-basin transfers may play an increasing role and could have substantial impact on larger rivers from which water is being transferred, as well as the receiving channel (Larned *et al.*, 2010b; Steward *et al.*, 2012).

Models and remote sensing are also frontiers in hydrologic sciences that have considerable potential when the cost for widespread, long-term monitoring becomes prohibitive. Field data are necessary to build and validate models and remote sensing. Two methods for future field research that we believe will help elucidate the complex interaction of meteorology, geology and land cover include (1) use of environmental tracers and isotopes for identifying age, sources and pathways of water within a riverscape that includes both the aquatic and associated terrestrial regions of a river network and (2) monitoring networks for assessing spatio-temporal patterns of flow within intermittent river networks. Meteorologic, geologic and land-cover watershed characteristics interact in complex ways, which influence flow pathways and how water is routed through landscapes. Isotopes (e.g. ^{18}O , ^2H and ^3H) provide information about transit time and environmental tracers (e.g. Si, Ca and F) provide information about the sources of water and flow pathways of hydrologic systems (McDonnell *et al.*, 2010). Multi-tracer approaches are necessary to fully characterize the sources and timing of water intermittent rivers. The use of environmental tracers and isotopes to assess hydrological characteristics of a system is particularly beneficial, requiring less *a priori* information, and can integrate heterogeneity within the system (Bracken *et al.*, 2013). Environmental tracers used in conjunction with isotopes are able to integrate small-scale hydrological processes and large-scale catchment responses (Soulsby *et al.*, 2006).

Traditional gages (e.g. located at a single station) are unlikely to capture the spatio-temporal variability of intermittent rivers (Snelder *et al.*, 2013). Field mapping of expansion and contraction has been conducted (Day, 1978; Day, 1980; Turner and Richter, 2011; Godsey and Kirchner, 2014), but the accuracy is limited by subjectivity of interpretation and the frequency of site visits. Electrical resistivity (ER) sensors are increasingly used to characterize the spatio-temporal dynamics of flow connectivity and continuity of intermittent rivers. ER sensors provide a low-cost, high-performance and minimal data interpretation method to monitor intermittent river spatio-temporal dynamics (Chapin *et al.*, 2014). Aspects other than timing, duration and frequency of flow such as the resumption (i.e. advancing) and recession (i.e. retreating) rates or magnitudes can be characterized with ER sensor arrays (Blasch *et al.*, 2002; Adams *et al.*, 2006; Goulsbra *et al.*, 2009; Bhamjee and Lindsay, 2010; Chapin *et al.*, 2014; Goulsbra *et al.*, 2014; Jaeger *et al.*, 2014; Bhamjee *et al.*, 2015; Peirce and Lindsay, 2015) or other instrumentation (e.g.

video, photographs, audio and temperature; Larned *et al.*, 2010a; Gungle, 2006; Constantz *et al.*, 2001; Spence and Mengistu, 2015; Quinlan *et al.*, 2015). A combined approach of instrumentation with piezometers to determine groundwater table elevations, soil water probes and ER sensor arrays would facilitate an understanding of expansion and contraction cycles at an unprecedented resolution over large spatial scales. However, ER sensors and piezometers alone are only able to determine presence/absence of water and cannot provide information about the source of water within a network, which would require tracer methods. Integrating piezometers, soil water probes and ER sensors remain intellectual and practical issues in translating massive datasets into scientific knowledge (Porter *et al.*, 2012). Using field-based measurements of flow permanence in conjunction with model simulations may provide valuable insight into how flow continuity and connectivity may change as a function of climatic shifts (Jaeger *et al.*, 2014).

Quantifying characteristics of streamflow continuity and connectivity are important for understanding ecological responses to this hydrologic disturbance. The degree of continuity will influence the ability of aquatic and riparian zone biota to endure and/or avoid a drying disturbance (i.e. resistance). Connectivity will influence the ability of biota dependent on water to recolonize from refuges (e.g. perennial reaches, groundwater and isolated pools) when water returns (i.e. resilience). Variability of flow permanence characteristics within and among networks may influence the harshness of meteorologic, geologic and land-cover conditions at local and regional scales, constraining the diversity of aquatic organisms that can persist at a given locality and the overall pool of species available to recolonize after drying. Hydrologic variation can be better characterized by using recently developed tools such as ER sensors that would enhance our ecological understanding of these transient systems and enable better predictions of ecological dynamics across a range of spatio-temporal scales.

Bridging disciplinary-specific methods and spatio-temporal scales of studies to produce a more consistent, synthetic understanding of intermittency is a primary challenge of studying intermittent rivers. Intermittent rivers are complex systems with various influences on the manner through which flow is distributed over time and space. Integrating environmental tracers, isotopes, piezometers and ER sensors provides an exciting opportunity to gain a comprehensive understanding of the timing, pathways and sources of water and the expansion and contraction cycles that are exemplified in intermittent rivers. If this integrated methodological approach is deployed long term with high spatio-temporal resolution, we can more thoroughly understand the meteorology, geology and land-cover characteristics that give rise to the presence and source of water within intermittent river networks.

CONCLUSIONS

The conceptual models and future research directions described in this paper are aimed at moving us towards a more comprehensive understanding of the complex hydromorphodynamics of intermittent rivers. Research focusing on intermittent rivers has increased rapidly with respect to the number of published studies and the maturity of the science within the last decade. We have classified research around intermittent rivers into three primary controls that represent distinct disciplines with discipline-specific concepts, methods and spatio-temporal scales of study. It is timely for researchers to begin unifying research on the meteorologic, geologic and land-cover aspects of intermittent rivers with biological studies of intermittent rivers across spatial and temporal scales that range from the watershed to sub-metre and millennial to hours.

Research design approaches that capture the structure as well as the functions that control the presence and absence of surface water and/or groundwater in intermittent river networks will contribute substantially to this field of study. Research integration from the fields of meteorology, geology and ecology will provide exciting opportunities for exploring the interactions of multiple spatio-temporal controls on intermittency and their implications for river ecosystems. An understanding of each scale of influence on the character of flow permanence provides a useful framework to appropriately describe flow permanence and better anticipate hydrological and ecological responses under changing climate change scenarios and/or water resources development. Coupling surface-water monitoring using ER sensors and piezometers with the use of environmental tracers and isotopes are necessary tools to assess intermittent rivers resulting in information gained that is a high research priority for the hydrological and ecological community. The temporal resolution and storage capacity of the latest generation of ER sensors enable long-term (i.e. multiple years) deployment and capture of network expansion and contraction dynamics. The conceptual models and future research directions described here are aimed at moving us towards a more comprehensive understanding of the complex hydromorphodynamics of intermittent river networks.

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